# Module lifecycle, packaging and billing boundaries: review and plan of action

**Prepared:** 8 September 2026. **Package:** MOD. **Status:** SOURCE_REVIEW_AND_FOCUSED_VERIFICATION; implementation not started by this run.

## 1. Assessment and scope

This is a source review and focused local verification, not a full runtime certification. Database clients in the selected tests are mocked or injected. No deployed database, authenticated browser, live provider, production telemetry, restore rehearsal or qualified statutory approval was inspected in this run. Unexecuted scenarios below are acceptance requirements, not test results. No application code, migration, business record or deployment is changed by this report.

**Accountable roles:** Platform engineering and commercial product operations. **Catalog coverage:** Cross-module workflow; not a standalone catalog promise.

**Selected verification:** 11 passed, 0 failed in the selected suite. Test source. This package owns or consumes PM-SCOPE-001.

The highest-risk review questions are catalog availability is not released capability; test direct services, jobs, APIs and routes under each entitlement mode; separate new-work dependencies from obligations that must remain completable.

## 2. Source and architecture map

The source inventory contains 39 matching files, including tests and supporting files. 5 sources were hashed and indexed for targeted review. Prefix inventory counts are discovery aids, not reviewed-file counts. The full list and source hashes are in the source manifest.

| Layer | Current source/navigation anchor | Evidence boundary |
|---|---|---|
| services | services/modules/module-entitlement.service.ts | 418 lines; SHA-256 e672098184a2… |
| actions | actions/modules/module-composition.actions.ts | 48 lines; SHA-256 365ceb8476c7… |
| actions | actions/modules/module-control.actions.ts | 96 lines; SHA-256 41cdb7cca286… |
| components | components/modules/TenantModuleActivationButton.tsx | 115 lines; SHA-256 5c7512eae51f… |
| components | components/modules/ModuleCompositionWorkbench.tsx | 590 lines; SHA-256 ca1e5fde1230… |

Connected route/API filenames in the inventory: app/[locale]/(dashboard)/dashboard/settings/modules/page.tsx.

## Data declarations

| Schema model | Selected declared storage/relationship facts |
|---|---|
| TenantModuleConfiguration | organization Organization @relation(fields: [organizationId], references: [id], onDelete: Restrict); @@map("tenant_module_configurations") |
| TenantModuleGrant | organization Organization @relation(fields: [organizationId], references: [id], onDelete: Restrict); configuration TenantModuleConfiguration @relation(fields: [organizationId], references: [organizationId], onDelete: Restrict, map: "tenant_module_grants_configuration_fkey"); @@unique([organizationId, moduleSlug]) |
| TenantModuleConfigurationCommand | organization Organization @relation(fields: [organizationId], references: [id], onDelete: Restrict); configuration TenantModuleConfiguration @relation(fields: [organizationId], references: [organizationId], onDelete: Restrict, map: "tenant_module_commands_configuration_fkey"); @@unique([organizationId, idempotencyKey], map: "tenant_module_commands_org_idempotency_key") |
| TenantModuleLifecycleOperation | configuration TenantModuleConfiguration @relation(fields: [organizationId], references: [organizationId], onDelete: Restrict, map: "module_lifecycle_configuration_fk"); command TenantModuleConfigurationCommand @relation(fields: [organizationId, commandId], references: [organizationId, id], onDelete: Restrict, map: "module_lifecycle_command_fk"); @@unique([organizationId, commandId]) |
| TenantModuleLifecycleStep | operation TenantModuleLifecycleOperation @relation(fields: [organizationId, operationId], references: [organizationId, id], onDelete: Restrict, map: "module_lifecycle_step_operation_fk"); @@unique([organizationId, operationId, moduleSlug, stepKey], map: "module_lifecycle_step_identity"); @@map("tenant_module_lifecycle_steps") |
| TenantModuleLifecycleEvent | operation TenantModuleLifecycleOperation @relation(fields: [organizationId, operationId], references: [organizationId, id], onDelete: Restrict, map: "module_lifecycle_event_operation_fk"); @@unique([organizationId, operationId, sequence], map: "module_lifecycle_event_sequence"); @@unique([organizationId, operationId, eventKey], map: "module_lifecycle_event_key") |

These are schema declarations. Exact migration application, actual constraints/indexes, row populations, query plans and direct-write bypasses remain unverified. See the platform package for migration/restore ownership.

## Workflow trace and handoffs

The diagram is the review boundary and required end-to-end trace. Source links and selected tests substantiate only the portions described in this report; it does not claim every edge was exercised.

```
flowchart LR
  A["Business trigger / source"] --> B["Fresh tenant and role authority"]
  B --> C["MOD: owning domain service"]
  C --> D["Transaction and source evidence"]
  D --> E["Event / dependent workflow"]
  D --> F["Read model / query scope"]
  F --> G["UI / export / recovery"]
  E --> H["Reconciliation / assurance"]
```

| Workflow | Expected outcome | Evidence disposition |
|---|---|---|
| MOD-FLOW-1 | Evaluate catalog, durable grant and permission authority on reads/writes | Source navigation and selected mocked/injected assertions only; full end-to-end scenario remains unverified |

| Workflow | Expected outcome | Evidence disposition |
|---|---|---|
| MOD-FLOW-2 | Stop new work while retaining mandatory settlement, corrections and historical access | Source navigation and selected mocked/injected assertions only; full end-to-end scenario remains unverified |
| MOD-FLOW-3 | Reconcile dependencies and commercial/provider changes without direct billing authority over business records | Source navigation and selected mocked/injected assertions only; full end-to-end scenario remains unverified |

**Integration contracts:** IAM — Identity, tenant access, settings and administration; EVT — Events, recovery, migrations and platform operations. These identify validation/ownership interfaces, not permission to duplicate their implementations. Accounting remains owned by the existing accounting roadmap.

**Graph navigation:** graph_actions.json node `modules_module_control_actions_ts`, community 116 (L1); graph_actions.json node `modules_module_control_actions_getmodulecontrolcenteraction`, community 116 (L57). Graph extracts predate this review and are not runtime proof.

# 3. Controls to preserve

The following assertions passed against mocks/injected clients in this run. Preserve their contracts while extending validation; test names are not broader certification claims.

- business new-work admission does not equate workspace omission with a revoked right.
- business new-work admission stops the selected capability and dependent new order creation.
- business new-work admission keeps mandatory ledger controls available when new manual journals stop.
- business new-work admission does not convert a recommended connection into an admission dependency.

# 4. Findings, limitations and risks

## PM-SCOPE-001 — Catalog entries and actual workflow evidence need reconciliation

**CAPABILITY_EVIDENCE_GAP · P2.** Confidence: High for catalog/search result; runtime/deployment scope unverified. Likelihood: Depends on which catalog promises reach users; not measured.

**Evidence:** services/modules/module-catalog.service.ts:243; line 171; line 255.

**Trigger:** A buyer or operator treats catalog status and route prefixes as proof of a complete workflow.

**Expected:** Every offered capability maps to a connected owner, route, data lifecycle and evidence-supported scope; aliases and deferred features are explicit.

**Observed:** Commercial Agents is available, Presence is available and Content is internal in the catalog. Exact path-name discovery found no presence/commercial-agents/blogs implementation family; HRIS attendance and public landing surfaces do exist elsewhere. This is a search result, not proof of complete absence.

**Cause / uncertainty:** Declared catalog scope and implementation naming/availability have not been reconciled in this review.

**Consequences:** False expectations, failed onboarding and entitlement confusion can damage trust. Commission liability or payroll scope must not be inferred from a catalog row.

**Existing mitigation:** Content is already marked internal; HRIS provides related workforce/time capability.

**Correction:** Trace aliases and deployment flags, then document supported scope or restrict misleading availability. Do not build speculative CMS/commission systems as part of hardening.

**Acceptance:** Every advertised catalog row has a verified destination and owner, or an explicit unavailable/internal disposition reflected in UI and commercial material.

**Owner / estimate:** Module platform and commercial product owners; 1–2 days of discovery; approved product changes separately estimated. Estimates are planning judgments, not booked capacity.

**Data and rollout:** No business-record changes; preserve existing grants and obligations when correcting availability.

**Related earlier work:** No earlier work ID assigned.

### Additional module scenarios still requiring verification

- **MOD-VERIFY-1:** Catalog availability is not released capability. Owner: Platform engineering and commercial product operations. Status: UNVERIFIED. Evidence required: a synthetic failure/success pair and source-to-result tie-out.
- **MOD-VERIFY-2:** Test direct services, jobs, APIs and routes under each entitlement mode. Owner: Platform engineering and commercial product operations. Status: UNVERIFIED. Evidence required: a synthetic failure/success pair and source-to-result tie-out.
- **MOD-VERIFY-3:** Separate new-work dependencies from obligations that must remain completable. Owner: Platform engineering and commercial product operations. Status: UNVERIFIED. Evidence required: a synthetic failure/success pair and source-to-result tie-out.

## 5. Layer and reviewer dispositions

| Lens | Current disposition and remaining proof |
|---|---|
| Architecture, backend and integrations | Sources/call boundaries inventoried; selected service assertions above. Trace alternate entry points and every handoff before whole-domain approval. |
| Database, migration and integrity | Named schema models indexed. Deployment, tenant-consistent foreign keys, locking, rollback, scale and backfill/restore not verified. |
| IAM, privacy, fraud and controls | Apply direct denied-no-read/no-write, role revocation, maker-checker and export redaction tests. A server guard or hidden button alone is insufficient. |
| Frontend, queries and design system | Client/action entry above inspected statically where present; cache identity, mutation effects and source truth must be traced into actual screens. |
| UX, accessibility, EN/FR and content | Authenticated role tasks, keyboard/focus, mobile and localized state behavior not executed; no accessibility claim. |
| Product and business process | Three explicit workflows and supported-scope decisions above; measure task completion and support burden before redesign. |
| Accounting and financial controls | Direct posting calculation is not applicable to this package; authority, claims and downstream financial effects remain in scope. |
| Statutory and country review | No new statutory interpretation is made. Escalate any discovered country-specific output to CMP/qualified review. |

| Lens | Current disposition and remaining proof |
|---|---|
| QA and release | 11 passed, 0 failed in the selected suite. linked test source. Unit results do not replace integrated evidence. |
| SRE, performance and cost | No latency/load/restore measurements; establish bounded reads, worker age/error metrics, alerts, retries and support ownership before rollout. |
| AI and agent authority | AI model behavior is not applicable unless a connected agent consumes this module; review that access contract without adding AI features. |
| Packaging, billing and operations | Confirm actual catalog/entitlement scope and mandatory retained obligations. Provider billing, adoption and customer outcomes are unverified. |

# 6. Ordered execution plan

## PM-MOD-01 — Bind current scope, owners and candidate

**Status:** READY. **Owner:** Platform engineering and commercial product operations. **Effort:** 1–2 reviewer days.

**Deliverable:** Confirm every listed workflow, supported variant and relevant prior ACC/ACR or STQ-RM item; hash current sources, map entry points and name a domain reviewer.

**Acceptance:** All material paths have active/legacy/deferred status; no existing change is overwritten.

**Prerequisite:** Candidate/owner availability; existing historical evidence is a lead..

## PM-MOD-02 — Reproduce priority risks and freeze contracts

**Status:** PLANNED. **Owner:** Platform engineering and commercial product operations. **Effort:** 2–4 reviewer/QA days; isolated environments may extend elapsed time.

**Deliverable:** Catalog availability is not released capability; Test direct services, jobs, APIs and routes under each entitlement mode; Separate new-work dependencies from obligations that must remain completable. Reproduce all linked findings before prescribing data repair.

**Acceptance:** A scenario ledger records expected/observed behavior, source IDs, denied side effects and remaining blockers.

**Prerequisite:** PM-MOD-01.

## PM-MOD-03 — Implement verified corrections and investigate affected data

**Status:** CONDITIONAL_ON_CONFIRMED_FINDINGS. **Owner:** Platform engineering and commercial product operations. **Effort:** Use linked finding estimates; otherwise estimate after diagnosis.

**Deliverable:** Correct the smallest owning boundary; preserve domain history, shared controls and public contracts. Investigate historical impact separately and design reversible repair only if needed.

**Acceptance:** Focused regression, cross-module tie-out and review prove the correction; no speculative implementation or silent historical rewrite.

**Prerequisite:** PM-MOD-02. If no defect is confirmed, record a reviewed no-change disposition; this satisfies the decision step without speculative work.

### PM-MOD-04 — Verify integrated release and controlled rollout

**Status:** PLANNED. **Owner:** Platform engineering and commercial product operations. **Effort:** 2–3 reviewer/QA days plus the real operational cycle.

**Deliverable:** Run isolated DB failure/concurrency scenarios, authenticated role/locale/cache tests, representative performance/export checks and recovery rehearsal. Pilot the agreed supported scope.

**Acceptance:** No unexplained material mismatch, unauthorized effect or stale evidence claim; domain reviewer and operations owner accept scoped results.

**Prerequisite:** PM-MOD-03.

Existing accounting and enterprise roadmap estimates must be deduplicated: PM findings linked to ACC/ACR or STQ-RM reuse the owning task. Reviewer-day estimates here are not engineering effort for a complete rewrite and must not be summed with the accounting roadmap as if independent.

## 7. Data correction, rollout and completion

Investigate exposure using source IDs, time/currency scope, counts and reconciled totals before proposing any repair. Preserve original records, hashes and exports; use reviewed corrective transactions or versioned migration only where necessary. Rehearse backup/restore and rollback in a disposable environment. Code correction and historical reconciliation are separate acceptance items. A query/display issue alone does not justify rewriting financial history.

Start with an agreed eligible pilot and observe a complete relevant cycle (sale/refund/shift; receiving/invoice/payment; payroll period; or close). Stop expansion on unauthorized effects, unexplained material mismatches, duplicate effects or stale proof. Record incident owner, recovery route and the data needed for investigation. No pilot or release is authorized or completed by this document.

Completion requires disposition for every workflow, current-source evidence for each significant claim, tested or explicit blocked failure scenarios, domain-reviewed corrective work and an understood operational handoff. The tailored audit prompt provides the full repeatable execution brief.

## 8. Evidence and provenance

Commit: `a05a41ef66cd5d98fd69ae3df18225795f912232` ; branch: `codex/migration-control-plane-release-2026-08-29` . Initial capture: 2026-09-08T14:41:48.494Z. Dirty workspace preserved; claims bind to file hashes, not clean HEAD. Unit command: 2026-09-08T14:43:08.844Z to 2026-09-08T14:44:14.458Z; exit 1. See exact command, assertion results, test log and shared evidence index. Historical reports were used for navigation and work deduplication; their earlier pass counts are not included in this run.